CARLOS BARRERA

Associate GIS Specialist | GIS Applications · Data Quality · GIS–IT Integration

Colorado Springs, CO · (719) 433-6284 · barrera.c1983@gmail.com · linkedin.com/in/carlos-barrera-gis
Portfolio: experience.arcgis.com/experience/3b769a75dd8440ec9f234fc76849ff1e

PROFESSIONAL PROFILE

GIS professional with 3 years of Esri experience, currently an Associate GIS Specialist (part-time) with El Paso County Digital Services & Technology. I build and maintain GIS applications, field workflows and county datasets, from requirements gathering through testing, publishing and user support. A background in network administration and telecommunications lets me work closely with IT staff. I use Python (ArcPy) and ArcGIS Notebooks to automate data preparation and publishing.

TECHNICAL SKILLS

GIS & Mapping: ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise / Portal, cartographic production, spatial analysis, georeferencing, remote sensing

Applications: Experience Builder, ArcGIS Dashboards, Field Maps, Survey123, Hub, web maps, hosted feature layers, Arcade

Data & Quality: file and enterprise geodatabases, schema development, attribute validation, QA/QC, topology review, metadata, photo attachments

Automation & Development: Python (ArcPy), ArcGIS Notebooks, ModelBuilder, REST services, testing with sample data

BIM & Indoor GIS: Revit, IFC, Dynamo

IT & Support: routers, switches and VLANs, software installation and troubleshooting, technical documentation, end-user support

PROFESSIONAL EXPERIENCE

Associate GIS Specialist | El Paso County — Digital Services & Technology Jan 2024 – Present
Colorado Springs, CO

- Maintain and update county GIS datasets, including parcels, jurisdictional boundaries, infrastructure and right-of-way; run QA/QC on geometry and attributes before data releases.
- Build, publish and maintain web maps, hosted feature layers, dashboards and Experience Builder applications in ArcGIS Pro, ArcGIS Online and ArcGIS Enterprise.
- **Affordable Housing Explorer:** worked with Economic Development on a public-facing housing application recognized with a 2024 County Innovation Award and viewed more than 30,000 times.
- **Homeless Camp Management (Code Enforcement):** built a Field Maps collection workflow with Arcade calculated fields, historical record tracking and a reporting dashboard; validated year, status, size and jurisdiction filters.
- **Data Communications (GIS + IT):** support fiber and conduit infrastructure mapping by reviewing geodatabase schemas, field requirements, asset attributes and contractor-provided data; flag records that need review.
- **Black Forest Water Sources (pilot):** organized cistern and pond records, matched source photographs to each site, and configured popups and attachments so fire districts can find water sources quickly.
- **Public safety:** produced hazardous-materials impact mapping and updated fire district applications for emergency response.
- Gather requirements from county departments and contractors, test with sample data before release, document workflows, and support users when issues come up.

- Georeference and align historical aerial imagery with current county GIS data; contributed to a County GIS Management Solution for request tracking and operational dashboards.

Network Administration Intern | University of Colorado Colorado Springs Jun 2022 – Jan 2024
Colorado Springs, CO

- Supported network and endpoint systems, including routers, switches and VLAN configurations; installed and troubleshot software, documented changes and resolved user issues.

Enterprise Solutions Technician | United Telephone Association Nov 2018 – May 2020
Dodge City, KS

- Supported business telecommunications and VoIP systems; configured equipment, diagnosed service problems and maintained technical documentation.

LIVE WEB MAPS & INDEPENDENT PROJECTS

- **BIM to GIS Indoor Map (2026):** modeled a two-floor building in Revit with a Dynamo Python script, exported it to IFC, and converted floors, rooms and network equipment into GIS layers with Python. Published as a web map with floor switching, room search, a 3D view and a network failure simulation, without an ArcGIS Indoors license.
- **Front Range Watch (2026):** tested 483 county planning projects against the wildland-urban interface and past burn scars; 49 sit in both. Includes an imagery swipe back to 2014 and live NASA fire detections.
- **EPC Live Conditions:** address lookup against FEMA flood zones, with live USGS stream gauges and NWS alerts pulled from public REST APIs.
- **GIS Data on Demand:** self-service tool that packages parcels, zoning and USGS 3DEP contours as Shapefile, CAD (DXF) or GeoJSON for a drawn area.
- **Pagosa Springs Flood (rapid response):** situation map for the Oct–Nov 2026 San Juan River floods with the full USGS gauge record and a dated event timeline.

Live links for each project are in the portfolio.

EDUCATION

University of Colorado Colorado Springs — Bachelor of Arts, Geography & Environmental Studies (in progress) · GIS Certificate (in progress)

Coursework in ArcGIS mapping, spatial analysis, remote sensing, cartography and geospatial data workflows.

MILITARY

U.S. Army — Non-Commissioned Officer. Led and trained soldiers and was responsible for equipment and operational readiness.